Emad Mahmodi

Senior Software Developer & Reverse Engineering Researcher
Expertise: Systems Security, AI in Cybersecurity, LLM/RAG for Binary Analysis

Research Interests

• AI & Cyber Security: LLM for Reverse Engineering, RAG for Firmware Documentation, AI-based Phishing Detection
• Data Science: Anomaly Detection in High-Frequency Streams, Trust Computation, Predictive Modeling
• Systems Security: Automotive ECU Architecture (Tricore/PowerPC), Vulnerability Research, Anti-debugging

Education

M.Sc. in Computer Science (Software Engineering)
Ferdowsi University of Mashhad, Iran | 2014 – 2017
GPA: 16.71/20 | Thesis: Adaptive rule base phishing detection against URL obfuscation (Grade: 18/20)

B.Sc. in Computer Science (Software Engineering)
Shomal University of Amol, Iran | 2008 – 2012
GPA (Last Year): 17.14/20 | Thesis: Integrated online sales management system (Grade: 20/20)

Publications

1. H. Koohi, Z. Kobti, T. Farzi, E. Mahmodi (2024). "UDIS: Enhancing Collaborative Filtering with Fusion of Dimensionality Reduction and Semantic Similarity." Electronics, 13(20), 4073. MDPI.

2. E. Mahmodi, H. S. Yazdi, A. G. Bafghi (2020). "A drift aware adaptive method based on minimum uncertainty for anomaly detection in data stream with skewed distribution." Expert Systems with Applications, Elsevier.

3. R. Nozari, R. B. Nozari, H. Koohi, E. Mahmodi (2020). "A Novel Trust Computation Method Based on User Ratings to Improve the Recommendation." International Journal of Engineering (IJE).

Professional Experience

Senior Software Developer & Reverse Engineering Researcher

• Leading R&D in AI-driven firmware analysis and automotive embedded security.
• Senior Developer for ECUProg v2: Architected a multi-protocol ECU programming suite

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(C++17/Qt 6) used in 5,000+ workshops. Achieved 49% less memory usage and 62% reduction in JTAG CPU load.
• Leading the Reverse Engineering of complex ECU architectures including TriCore (Infineon) and PowerPC (VLE) for major automotive brands (Lamari, Bajaj, etc.).
• Developed custom Ghidra and Radare2 extensions in Java/Python for binary de-obfuscation and automated ISA analysis.
• Implemented high-speed diagnostic gateways using Service Oriented Architecture (SOA) and kernel-bypass techniques.

Senior System Programmer | NIUniverse & Rahgozin Rayaneh
2020 – 2022
• Specialized in binary security auditing, malware analysis, and protocol simulation.
• Conducted deep-level vulnerability assessments of diagnostic software layers, focusing on anti-debugging and V-table obfuscation.

Academic Researcher & Software Consultant | Freelance
2017 – 2020
• Developed machine learning models for anomaly detection and phishing prevention as part of research publications.
• Consulted on network security and distributed deep learning projects (BiLSTM/Sentiment Analysis).

Certificates & Honors

• Reviewer Certificate: 10th International Conference on Computer and Knowledge Engineering (ICCKE 2020), Ferdowsi University of Mashhad (Acknowledged for reviewing 4 papers).
• Teaching Certification: Expert Instructor in Data Mining and Machine Learning.
• Academic Advising: Certified M.Sc. Thesis Advisor / Research Consultant.

Key Technical Projects

• Locomotive GT26 Control System: Implemented compile-time/run-time State Machines (Simulink-like) in C++ for railway infrastructure. Developed FPGA-to-PC interfaces via Verilog and Shared Memory.

• PISAD ECU Full Simulator: Built a hardware-level PowerPC emulator for firmware security auditing and fault-injection research.

• LLM-Assisted Reverse Engineering: Integrating LLMs with binary analysis tools for automated code summarization and documentation lookup using RAG.

Technical Skills

• AI/ML: LLM Fine-tuning, RAG, PyTorch, Anomaly Detection, Data Science.
• Engineering: C++17/20, Qt/QML, Python, Java, Verilog, SIMD, OpenCL.
• Security: Ghidra, IDA Pro, Binary Analysis, UDS/CAN-Bus, Anti-tamper.

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